

```

1  #include <stdio.h>
2  int main() {
3      int i, n, time = 0, remain, tq;
4      int at[10], bt[10], temp[10];
5      int wt = 0, tat = 0;
6      printf("Enter total number of processes: ");
7      scanf("%d", &n);
8      remain = n;
9      for(i = 0; i < n; i++) {
10         printf("\nEnter Arrival Time and Burst Time for Process %d:\n", i + 1);
11         printf("Arrival Time: ");
12         scanf("%d", &at[i]);
13         printf("Burst Time: ");
14         scanf("%d", &bt[i]);
15         temp[i] = bt[i]; // Copy burst time
16     }
17     printf("\nEnter Time Quantum: ");
18     scanf("%d", &tq);
19     printf("\nProcess\tBurst Time\tTurnaround Time\tWaiting Time\n");
20     int count = 0;
21     for(i = 0; remain != 0; ) {
22         if(temp[i] <= tq && temp[i] > 0) {
23             time += temp[i];
24             temp[i] = 0;
25             count = 1;
26         }
27         else if(temp[i] > 0) {
28             temp[i] -= tq;
29             time += tq;
30         }
31         if(temp[i] == 0 && count == 1) {
32             remain--;
33             int turnaround = time - at[i];
34             int waiting = turnaround - bt[i];
35             printf("P[%d]\t%d\t\t%d\t\t%d\n", i+1, bt[i], turnaround, waiting);
36             wt += waiting;
37             tat += turnaround;
38             count = 0;
39         }
40         if(i == n - 1)
41             i = 0;
42         else if(at[i + 1] <= time)
43             i++;
44         else
45             i = 0;
46     }
47     printf("\nAverage Turnaround Time = %.2f", (float)tat / n);
48     printf("\nAverage Waiting Time = %.2f\n", (float)wt / n);
49     return 0;
50 }

```

Total number of process in the system: 4

Enter the Arrival and Burst time of the Process[1]

Arrival time is: 1

Burst time is: 23

Enter the Arrival and Burst time of the Process[2]

Arrival time is: 2

Burst time is: 32

Enter the Arrival and Burst time of the Process[3]

Arrival time is: 3

Burst time is: 2

Enter the Arrival and Burst time of the Process[4]

Arrival time is: 4

Burst time is: 45

Enter the Time Quantum for the process: 5

| Process No    | Burst Time | TAT | Waiting Time |
|---------------|------------|-----|--------------|
| Process No[3] | 2          | 9   | 7            |
| Process No[1] | 23         | 64  | 41           |
| Process No[2] | 32         | 85  | 53           |
| Process No[4] | 45         | 98  | 53           |

Average Turn Around Time: 38.500000

Average Waiting Time: 64.000000